

The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Control

Exact Bayes envelope generally; mutual-information bound universally;
closed form in the symmetric Markov case

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Abstract

A controller that acts on information of age D — a sensing delay, a telemetry lag, a stale cache — can outperform the best static policy only to the extent that the delayed observation still *predicts* the current state. Our companion systems work [1] proves this exactly for symmetric two-state Markov capacity ($\Delta(D) = \frac{1}{2}r(D)$, the autocorrelation at the sensing lag) with a β -mixing converse. This paper supplies the general theory. We (i) isolate the exact object — the *Bayes envelope* $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, the expected excess value of posterior over prior beliefs; (ii) prove the universal *predictive-information envelope* $0 \leq \Delta_Y \leq \Omega\sqrt{\frac{1}{2}I(X;Y)}$ for any bounded objective with oscillation Ω , by Pinsker’s inequality applied to the posterior process; (iii) derive the *bit-budget corollary*: a telemetry path delivering b bits per decision caps feedback value at $\Omega\sqrt{(b\ln 2)/2}$ regardless of what is encoded — compression can preserve but never create feedback value; (iv) show the envelope is *tight to leading order* in the canonical case: for the symmetric binary chain $I = \frac{r^2}{2} + O(r^4)$, so the bound reads $\Omega(\frac{r}{2} + O(r^3))$ against the exact $\frac{1}{2}r$; and (v) prove by a two-bit construction that no universal equality $\Delta = f(I)$ can hold: mutual information is a scalar, decision value is a geometry, and two observations with identical I can carry all or none of the decision-relevant distinction. A five-domain empirical section (cluster contention, acoustic, seismic, space weather, bioacoustic) tests the law and its envelopes on real processes.

1 Introduction

[TODO: motivation prose; position vs. AoI (freshness metrics without decision semantics), value-of-information (Howard [2]; De Lara–Gossner [3]), remote estimation, predictive information (Bialek et al.), delayed-sharing structures (Witsenhausen). The one-line thesis: *age matters only through predictive information, and predictive information matters only through the Bayes envelope.*]

2 Setting and the exact object

Let X be the current exogenous state taking values in a finite set $\mathcal{X}$, with marginal $\bar{\mu}$. A controller observes a delayed quantity Y (a delayed sample X_{t-D} , a delayed history, or any $\sigma(\mathcal{F}_{\leq t-D})$ -measurable statistic), chooses an action w , and receives a bounded one-step reward $R(w, x)$ with

oscillation $\Omega = \max_{w,x} R - \min_{w,x} R$. Define the value of a belief $\mu \in \Delta(\mathcal{X})$:

$$\psi(\mu) = \max_w \sum_x R(w, x) \mu(x),$$

a pointwise maximum of linear functionals, hence convex; and the posterior $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$, with $\mathbb{E}[\mu_Y] = \bar{\mu}$ by the tower property. The optimal Y -measurable controller attains $V_Y^* = \mathbb{E}[\psi(\mu_Y)]$; the best static controller attains $V_{\text{stat}} = \psi(\bar{\mu})$. The exact object of this paper is the **Bayes envelope**

$$\Delta_Y = V_Y^* - V_{\text{stat}} = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu}) \geq 0, \quad (1)$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

Lemma 1 (Lipschitz value [1, App. A]). $|\psi(\mu) - \psi(\mu')| \leq \Omega \|\mu - \mu'\|_{\text{TV}}$, hence $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$.

The total-variation route specializes to the β -mixing converse of [1] when Y is the full delayed history: $\Delta(D) \leq \Omega \beta(D)$.

3 The predictive-information envelope

Theorem 2 (Predictive-information bound). *For any bounded objective with oscillation Ω and any delayed observable Y ,*

$$0 \leq \Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Y)} \quad (I \text{ in nats; in bits, } \Omega \sqrt{\frac{\ln 2}{2} I_{\text{bits}}}). \quad (2)$$

Proof. By Lemma 1, $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$. Pinsker's inequality gives, for every realization y , $\|\mu_y - \bar{\mu}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_y \parallel \bar{\mu})}$. Taking the expectation over Y and applying Jensen to the concave square root,

$$\mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}} \leq \mathbb{E} \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_Y \parallel \bar{\mu})} \leq \sqrt{\frac{1}{2} \mathbb{E} D_{\text{KL}}(\mu_Y \parallel \bar{\mu})} = \sqrt{\frac{1}{2} I(X; Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior. $\square$

Corollary 3 (Bit-limited telemetry). *If the sensing path delivers at most b bits per decision — $Y = g(Z)$ with Z the raw delayed signal and $H(Y) \leq b \ln 2$ nats — then $\Delta_Y \leq \Omega \sqrt{(b \ln 2)/2}$, and moreover $\Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Z)}$ regardless of g (data processing). Compression can preserve feedback value; no encoding can create it.*

Proposition 4 (Leading-order tightness in the canonical case). *For the symmetric two-state Markov chain with lag- D autocorrelation $r = r(D)$,*

$$I(X_t; X_{t-D}) = \frac{1}{2} [(1+r) \ln(1+r) + (1-r) \ln(1-r)] = \frac{r^2}{2} + \frac{r^4}{12} + O(r^6),$$

so the right side of (2) equals $\Omega(\frac{r}{2} + O(r^3))$, matching the exact law $\Delta(D) = \frac{1}{2} r(D)$ of [1] to leading order (at the objective normalization $\Omega = 1$). The information envelope is not loose where it matters.

Proof. The joint distribution places mass $\frac{1+r}{4}$ on each agreeing pair and $\frac{1-r}{4}$ on each disagreeing pair; the stated I follows by direct computation, and the expansion from $(1 \pm r) \ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \dots$. Then $\sqrt{\frac{1}{2} I} = \frac{r}{2} \sqrt{1 + \frac{r^2}{6} + O(r^4)}$. $\square$

4 No universal equality $\Delta = f(I)$

Proposition 5 (Information is a scalar; value is a geometry). *There is a state space, a bounded objective, and two observations Y_1, Y_2 with $I(X; Y_1) = I(X; Y_2) > 0$ but $\Delta_{Y_1} = \frac{\Omega}{2} \cdot \frac{1}{2} > 0 = \Delta_{Y_2}$. Hence no function f with $\Delta_Y = f(I(X; Y))$ exists for all experiments.*

Proof. Let $X = (B_1, B_2)$ be two independent fair bits and let the objective depend only on B_1 : actions $w \in \{0, 1\}$, $R(w, x) = \Omega \mathbf{1}[w = B_1]$ (so the oscillation is Ω). Static value: $\Omega/2$. Let $Y_1 = B_1$ and $Y_2 = B_2$. Both have $I(X; Y_i) = \ln 2$. Observing Y_1 yields $V^* = \Omega$, so $\Delta_{Y_1} = \Omega/2$; observing Y_2 leaves the posterior over B_1 uniform, so $\Delta_{Y_2} = 0$. The two experiments share every information quantity in $I(X; Y)$ yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other. $\square$

Remark 6. *This is the precise sense in which the companion paper’s binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains (1).*

5 Envelope comparison

[TODO: β -mixing vs. $\sqrt{I/2}$: incomparable in general (examples both ways); both $\sim r/2$ for the binary chain; when Y is the full history, I may exceed the single-sample information while β saturates. Table of the three envelopes (TV/ β , MI, bit-budget) with their hypotheses and regimes.]

6 Exactness beyond two states

[TODO: layer-cake lift (cite [1, Prop. 2]): exact $\Delta(D) = \sum_u \pi_u r_u(D)$ for monotone-threshold objectives — already a multi-state exact law; characterize the objective class for which the Bayes envelope admits closed form; asymmetric two-state case.]

7 The law across domains

[TODO: full five-domain battery (cluster contention, LOCATA acoustic, PDAR seismic, GOES space weather, sperm-whale codas): measured $\Delta(D)$ vs. $\frac{1}{2}r(D)$ vs. the MI envelope; Markov-surrogate calibration; the INFOCOM paper carries one paragraph of this — the full protocol, controls, and per-domain figures belong here. Data: /archive/infocom.battery (Atlas).]

8 Related work

[TODO: Howard VoI [2]; De Lara–Gossner [3]; AoI surveys; remote estimation / Witsenhausen delayed sharing; Bialek predictive information; rate-distortion with side information; NCS.]

9 Conclusion

[TODO:]

References

- [1] A. H. Bond. Polite bursting: the value of delayed feedback in age-limited admission control. Under submission, IEEE INFOCOM 2027.
- [2] R. A. Howard. Information value theory. *IEEE Trans. Systems Science and Cybernetics* 2(1):22–26, 1966.
- [3] M. De Lara, O. Gossner. Payoffs-beliefs duality and the value of information. *SIAM J. Optimization* 30(1):287–310, 2020.